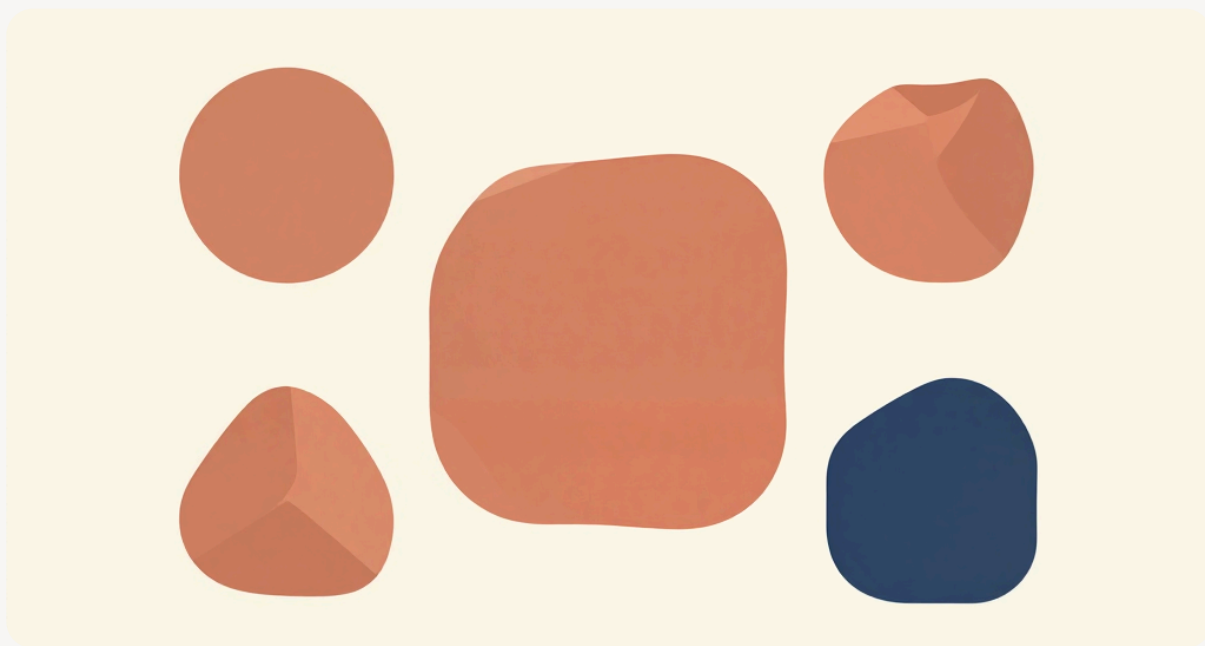




PAPER EXPLAINED

LLM agents fail at planning: here is the evaluation landscape

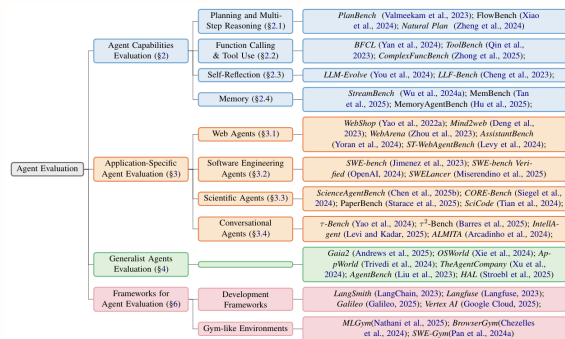
Survey on Evaluation of LLM-based Agents

Asaf Yehudai, Lilach Eden, Alan Li, Guy Uziel et al.

<https://arxiv.org/abs/2503.16416>

Mapping LLM Agent Evaluation Landscape

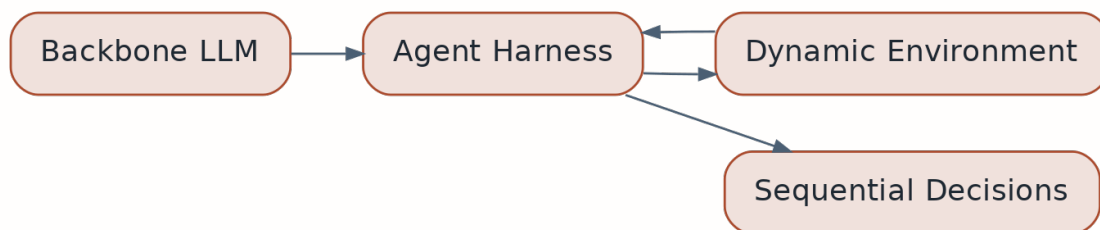
- LLM-based agents integrate models into multi-step workflows with external tools for autonomous planning.
- This survey presents the first overview of LLM-based agent evaluation by mapping the current landscape.
- The survey covers fundamental capabilities like planning, tool use, self-reflection, and memory.
- It reviews benchmarks for web, software engineering, scientific, and conversational agents.



Overview of the paper with core works.

Why Evaluating Agents Is Hard

- Evaluation must go beyond textual outputs to assess sequential decision-making in dynamic environments.
- Benchmarks reveal current methods struggle with long-range consistency and dynamic memory handling.
- Recent work exhibits over-optimistic estimates, proposing Online-Mind2Web as a more rigorous alternative.
- Most benchmarks conflate LLM capabilities with agent harness design, obscuring performance attribution.



Agent evaluation conflates LLM capabilities with harness design in dynamic environments.

Benchmarks: WebArena, SWE-bench, Gaia

- WebArena provides dynamic environments with fully functional websites across multiple domains.
- Gaia requires reasoning, multi-modality, web browsing, and tool-use on real-world questions.

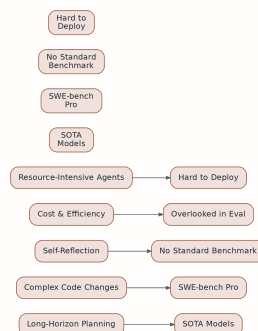
Gaia
Real-world Questions
Reasoning & Tool-use

WebArena
Dynamic Environments
Multi-domain Websites

Key benchmarks: WebArena and Gaia for evaluating LLM agents.

Performance Gaps and Critical Limitations

- State-of-the-art models struggle with long-horizon planning tasks.
- SWE-bench Pro performance remains below 25% Pass@1 for complex code changes.
- No standardized benchmark exists for assessing self-reflection capabilities.
- Evaluations overlook cost, creating resource-intensive agents hard to deploy.



Critical limitations in current LLM agent evaluation.